

WASP-43b

R-1066

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160318 · created 2026-07-30T12:54:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457465.7486 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.021
Transit depth (Rp/Rs)^2	2.82 +/- 0.71 [%]
Orbital Inclination (inc)	80.56 +/- 1.87
Airmass coefficient 1 (a1)	1.092 +/- 0.013
Airmass coefficient 2 (a2)	-0.06 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	None
Optimal Aperture	3.33
Optimal Annulus	6.52
Transit Duration (day)	0.0443 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.021 vs 0.1594 (deviation 0.35 σ)
Duration fit vs published	0.0443 d vs 0.0512 d (deviation 0.62 σ)
Mid-time O-C	-2.7 min
Depth SNR	6.9
Residual scatter	1.19 %

Light curve

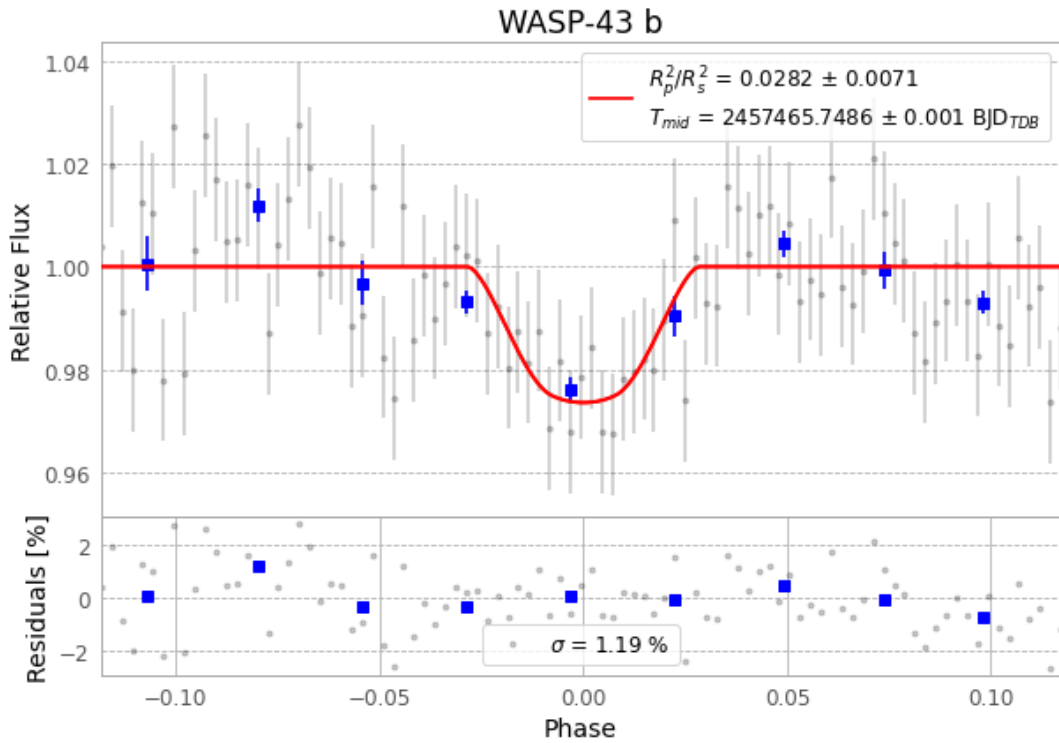


Figure 1 — FinalLightCurve_WASP-43 b_2016-03-17.png (SHA-256 729e3e362dfd29a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [104, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9d7374ffb99a46c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), WASP-43160318033012.FITS → WASP-43160318080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160318.log (97f68b6e4408...), FinalLightCurve_WASP-43 b_2016-03-17.png (729e3e362dfd...), FinalLightCurve_WASP-43 b_2016-03-17.csv (53e505a708b0...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 12:54Z.